

NATIONAL INSTITUTE OF TECHNOLOGY ROURKELA
Department Of Computer Science & Engineering
Supplementary Examination, Summer Semester 2023
B.Tech. 6th Semester

Subject Name: Computer Networks

Maximum Marks: 50

Subject Code: CS3002

Exam Duration: 3 Hours

Number of pages:1

Note: There are Four questions. Attempt All the questions. All the parts of every question must be attempted in a sequence.

Q.1 Attempt any Four from the following:

- (a) Explain the TCP/IP protocol suits by indicating the functionalities at each layer.
- (b) Describe the different types of delays that occurred during communication.
- (c) What do you understand by HTTP? Differentiate between Nonpersistent and Persistent HTTP. Also compare their response time in terms of RTT and transmission time.
- (d) Compare the Iterative and Recursive query used in DNS protocol in terms of latency.
- (e) Describe the TCP header information by showing length of its in bits/Bytes.
- (f) What is the use of CRC protocol. Explain with the help of example.

[4*5=20 Marks]

Q.2(a.) Explain the Slow Start and Additive Increase Multiplicative Decrease (AIMD) mechanism with the help of diagram.

(b.) What is CSMA/CD technique? Where is it used? Consider the Ethernet which uses CSMA/CD. What is the probability that Ethernet (Adaptor) would be transmitting the frame after 3rd subsequent collision.

[5+5=10 Marks]

Q.3 (a.) Differentiate between Distance Vector (DV) and Link State (LS) routing algorithms.

(b.) Suppose that host A is connected to a router R 1, R 1 is connected to another router, R 2, and R 2 is connected to host B. Suppose that a TCP message that contains 912 bytes of data and 20 bytes of TCP header is passed to the IP code at host A for delivery to B. Show the Total length, Identification, DF, MF, and Fragment offset fields of the IP header in each packet transmitted over the three links. Assume that link A-R1 can support a maximum frame size of 1024 bytes including a 14-byte frame header, link R1-R2 can support a maximum frame size of 512 bytes, including an 8-byte frame header, and link R2-B can support a maximum frame size of 512 bytes including a 12-byte frame header.

[4+6=10 Marks]

Q.4 Discuss any Two protocols from the following:

- (a.) Internet Protocol (IP)
- (b.) Aloha Protocol
- (c.) Dynamic Host Configuration Protocol (DHCP)
- (d.) Go-Back-N (GBN) Automatic Repeat Request (ARQ) Protocol

[5+5=10 Marks]